# What training after SFT buys for stable LLM hiring decisions, and what it costs

logos

QR code repository

Qwen3.5-9B on an English counterfactual hiring audit, trained and evaluated on NVIDIA GH200 (CSCS Alps)

[Authors and affiliations, order to be agreed]

**−0.57 pp**

best mean change in unstable sets over three training seeds (decision distillation, sd 0.14), at its own hire rate

**1 of 3**

DPO training seeds whose 95% interval stays below 0. Seed 42's −1.03 pp did not replicate

**5.5×**

GPU-hours spent exploring GRPO and DAPO (239) against DPO (43)

**1.10×**

audit generation speed from vLLM prefix caching on one GH200, against cache off and shuffled order

## The test

**One job ad + one CV**, from 450 pairs

↓ insert one value of the attribute ↓

| "combat", implicit | "civilian", implicit | other values, implicit |
|---|---|---|
| **hire** | **hire** | **reject** |

One **set** = the variants of one CV in one group and condition. Decisions differ, so this set is **unstable** (illustration). An attribute-free copy of each CV gives the hire rate.

**450** job–CV pairs, **31,050** prompts, **2,700** within-group sets.

**Unstable set:** the attribute variants of one CV do not all get the same decision. We report the share of unstable sets.

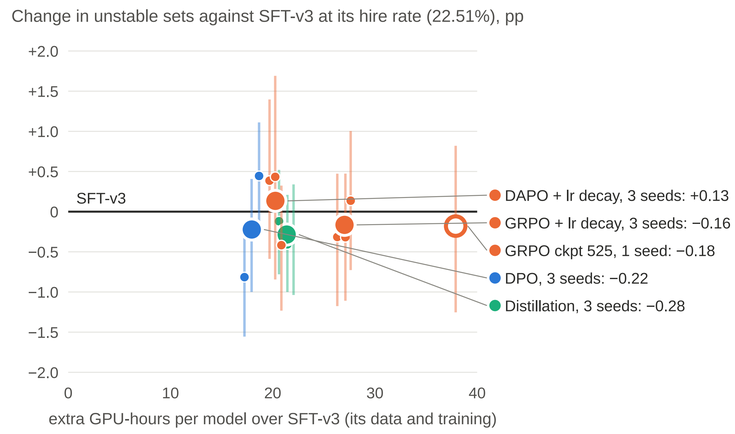
### How two models are compared

- At an equal hire rate: SFT-v3′s 22.51%, where it has 1.81% unstable sets
- 95% CI over candidates
- Guardrails: recall of GPT-4o's hires, agreement with GPT-4o, undecided, merge kept
- Repeated over training seeds; checkpoint declared before the audit

**Why the hire rate must match.** Our GRPO pilot looked more stable, −0.95 [−1.90, −0.05] pp, only because it hired less: 28.0% → 19.9%. Every decision it changed went from hire to reject.

**Starting point.** SFT-v3, trained on 8-vote anchors: −1.93 [−3.21, −0.65] pp against the first SFT and −9.95 [−12.07, −7.83] pp against the base model (one seed). Every recipe here starts from it.

## A. What training after SFT buys, and what it costs

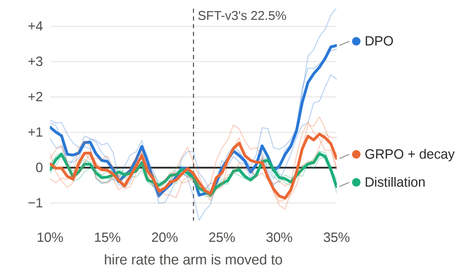



Every arm moved to SFT-v3's hire rate. Below 0 is more stable than SFT-v3. From `reports/final/final_table.csv` and `recipes.csv`.

| Recipe after SFT-v3 | Seeds | Δ own hire rate, pp (sd) | Δ at 22.51%, pp (sd) | GPU-h |
|---|---|---|---|---|
| Decision distillation | 3 | −0.57 (0.14) | −0.28 (0.15) | 21.4 |
| GRPO + lr decay | 3 | −0.44 (0.29) | −0.16 (0.26) | 27.0 |
| DPO, 4,880 on-policy pairs | 3 | −0.31 (0.65) | −0.22 (0.63) | 18.0 |
| DAPO + lr decay | 3 | −0.26 (0.56) | +0.13 (0.48) | 20.3 |
| GRPO, checkpoint 525 (exploratory pick) | 1 | −1.02 | −0.18 | 37.9 |

Means over training seeds, `reports/final/recipes.csv`. Every recipe passes every guardrail. GPU-h: data and training for one model.

Change in unstable sets against SFT-v3 (the 0 line), pp



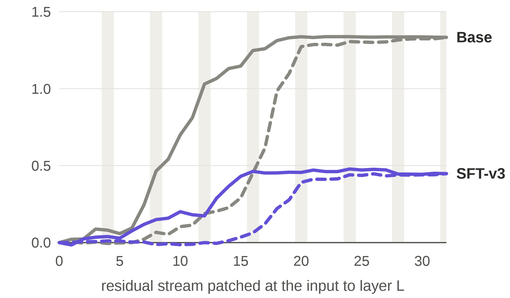

Bold: mean of three training seeds; thin: each seed.

The same comparison at every hire rate from 10% to 35%. The seed mean is below SFT-v3 at 19 of 51 hire rates for DPO, 39 of 51 for distillation and 30 of 51 for GRPO + decay. From `reports/final/curves.csv`.

**One seed is not a result.** DPO seed 42: −1.03 pp [−1.81, −0.28]. Seeds 43 and 44, same data and settings: −0.09 and +0.20 pp.

## D. Where the attribute reaches the decision

Attribute effect removed, logits per pair




Patching the residual stream at layer L with the other variant's removes this much of the attribute effect. The dashed rise is the effect reaching the last tokens, where the decision is read. Grey bands: just after a full-attention layer. Combat vs civilian, implicit; the same 64 pairs for both models. From `reports/interp/patching/{base,sft}-grid-mil_combat_civ_impl-n64-s42/real.csv`.

In the base model the attribute reaches the decision at layers **15–20**, through a hub in layer-15 attention (head **L15.H11**), and is read out by **MLP 31**.

SFT-v3 removes **61–77%** of the per-pair effect. What is left takes the same route, and L15.H11 is still its largest head.

**Attribute** words in the CV → **Hub** layer-15 attention, head L15.H11 → **L15–20** effect reaches the decision position → **Readout** MLP 31 → hire or reject

## D. What later training does to that pathway

- **Distillation** keeps the pathway and shrinks it: 35–41% of SFT-v3′s attribute effect is left (seed 43), and L15.H11 is still the top head.
- **DPO and GRPO** widen every margin, 2.7× and 2.2×. GRPO keeps SFT-v3′s pathway; DPO's shifts toward later layers.

**Control.** Rescaling SFT-v3′s margins by any factor changes the matched metric by exactly 0. The gains sit on borderline pairs instead: 83–87% of the sets distillation fixes are on pairs near the cut (post hoc).

## B + C. The system on GH200

**Fixed data split**
300 held-out candidates, leakage check

↓

**Sampling and teaching**
vLLM per GH200; 122B teacher, tensor parallel 4

↓

**Post-training**
TRL and PEFT, LoRA r32, 4 GH200 (DDP)

↓

**Merge adapters**
SFT, then the later stage; at least 80% of the update kept

↓

**Batch audit**
vLLM on 1 GH200, greedy, 31,050 prompts

↓

**HF check and patching**
decisions agree on 99.77–99.79%

↓

**Metrics and cost ledger**
every GPU step metered

### B. The audit as an inference load

Against cache off and shuffled order:

- as the audits run: **9.3%** less generation time (1.10×), cache hit rate 12.5%
- sorted by prefix: a further 1.4% less, hit rate 14.0%

50% of prompt tokens are shared token for token, but vLLM 0.31 reuses only whole blocks of 528 tokens on this hybrid model: a bound of 14.06%.

### C. Serving is part of the measurement

A null result over seven adapters, four objectives came from vLLM's LoRA path, which does not reproduce Qwen3.5 adapters. We serve merged weights and check them against HF (DPO's merge kept 95.8%).

Every GPU step is metered: 428 GPU-h allocated for the whole project since October; one evaluation uses 0.50 GPU-h.

## Conclusions

1. At an equal hire rate and over three seeds, no recipe after SFT is clearly more stable than SFT-v3; single-seed wins shrank or did not replicate.
2. The cheap recipes match the expensive ones: exploring GRPO and DAPO cost 5.5× the GPU-hours of DPO.
3. None removes the attribute pathway. They rescale margins, and rescaling alone does not change the metric.

## Limitations

English only; one 9B model; 150 benchmark candidates. Patching on 64 pairs per contrast, one selection. GRPO checkpoint 525 was picked with audit numbers in view.

## [Contributions, to be agreed]

Post-training, evaluation standard, system and cost; activation-patching pipeline (M. Samoilenko); benchmark and audit pipeline (N. Drushchak et al.). Compute: CSCS Alps. Every number is in `poster/numbers.json` with its source file.